**REST APIs and API Integration**

**1. Introduction to REST APIs**

REST (Representational State Transfer) is an architectural style used to build web services that allow different applications to communicate with each other over the internet. REST APIs commonly use HTTP and exchange data in formats such as JSON.

REST APIs are widely used because they are simple, flexible, scalable, and easy to integrate with web and mobile applications.

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**2. Importance of REST APIs**

REST APIs are important in modern applications because they:

- Allow different applications and services to communicate.
- Make data available to multiple platforms.
- Support scalable application development.
- Reduce the need to duplicate data and functionality.
- Can be used with web, mobile, and desktop applications.
- Provide a standard way of exchanging information.

For example, a weather application can request weather information from a weather API instead of maintaining its own weather database.

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**3. How REST APIs Work**

A REST API follows a client-server model.

```
Client Application
    |
    | HTTP Request
    ↓
  REST API
    |
    | JSON Response
    ↓
Client Application
```

The client sends a request to an API endpoint. The server processes the request and returns a response, usually containing JSON data.

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### 4. HTTP Methods

REST APIs commonly use the following HTTP methods:

| Method | Purpose |
|---|---|
| GET | Retrieve data |
| POST | Create new data |
| PUT | Update existing data |
| DELETE | Delete data |

For example:

GET /users

can be used to retrieve a list of users.

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### 5. JSON Responses

JSON (JavaScript Object Notation) is one of the most commonly used formats for API responses.

Example:

```
{
  "city": "Mumbai",
  "temperature": 28,
  "humidity": 75
}
```

JavaScript can convert the API response into an object using:

```
const response = await fetch("API_URL");
const data = await response.json();

console.log(data);
```

The json() method converts the response into a JavaScript object that can be used by the application.

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### 6. Making a GET Request

A GET request can be made using JavaScript's fetch() function.

```
async function getData() {
  try {
    const response = await fetch("https://example.com/api/data");
```

```
        const data = await response.json();

        console.log(data);
    } catch (error) {
        console.log("Error:", error);
    }
}
```

Here, fetch() sends the request, json() processes the response, and try...catch handles possible errors.

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**7. Integrating a Weather API**

For the practical implementation, a weather API can be used to retrieve current weather information based on a user's selected location.

The basic process is:

Enter City

↓

Send API Request

↓

Receive JSON Response

↓

Extract Weather Information

↓

Display Data in React

Example:

```
const response = await fetch(
    "https://api.open-meteo.com/v1/forecast?latitude=19.07&longitude=72.87&current=temperature_2m"
);

const data = await response.json();

console.log(data.current.temperature_2m);
```

The application can then display the temperature obtained from the API.

**8. Error and Response Handling**

API requests may fail because of network problems, incorrect URLs, or unavailable data. Therefore, applications should handle errors properly.

```
try {
   const response = await fetch("API_URL");


   if (!response.ok) {
      throw new Error("Failed to fetch data");
   }


   const data = await response.json();


   console.log(data);
} catch (error) {
   console.log(error.message);
}
```

This makes the application more reliable and provides better feedback to users.

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**9. Practical Application**

A **React Weather Application** was created as the practical implementation of REST API integration.

The application:

- Accepts a city name from the user.
- Sends requests to a public weather API.
- Receives weather information in JSON format.
- Processes the response using JavaScript.
- Displays temperature, humidity, and wind speed.
- Handles loading and error conditions.
- Uses responsive design for different screen sizes.

**Project Link:**
https://github.com/VedantNaik143/MERN-FSD/tree/0817bbbc4864f32d4c763722de0a2d4f7443f236/react-weather-app

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**10. Benefits of REST APIs**

Some major benefits of REST APIs are:

- Simple and easy to understand.
- Platform independent.
- Supports multiple data formats.
- Easy integration with third-party services.
- Scalable for large applications.
- Reduces development time by reusing existing services.

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**11. Industry Applications**

REST APIs are used extensively in real-world applications.

**E-commerce:** APIs connect products, payments, orders, and customer services.

**Social Media:** APIs allow applications to retrieve posts, profiles, comments, and other information.

**Banking:** APIs are used for transactions, account information, and payment services.

**Maps and Transportation:** Applications use APIs to obtain location, route, traffic, and transportation information.

**Weather Applications:** Weather APIs provide real-time weather information to applications without requiring them to maintain their own weather data.